Schematic Sheets

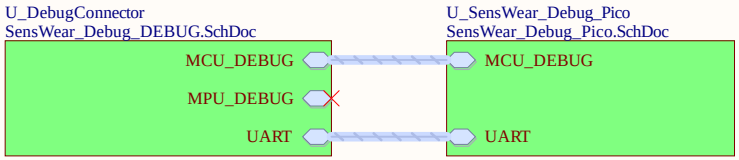
#	Name	File	Description
1	Main	SensWear_Debug_Main	Block Interconnect
2	DEBUG Connector	SensWear_Debug_DEBUG	Debug Connector
3	Pico Debugger	SensWear_Debug_Pico	Pico Debugger

Revision History

Revision	Date	Description
R0	June 2026	Initial validation design
R1	August 2026	FPF1320 current leakage to solve robust TVDD selection. MOSFET is updated to one that has lower VGS threshold

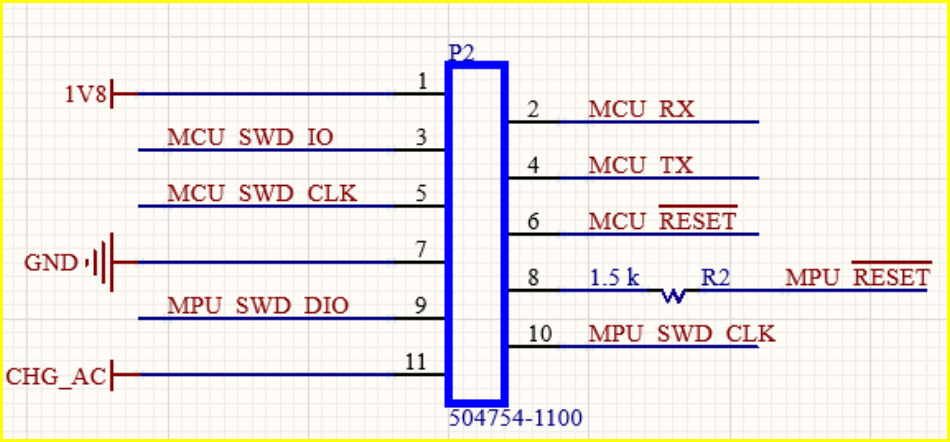
SenseWare-V1

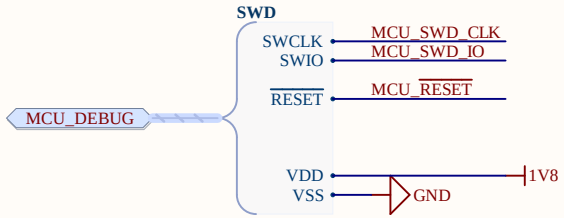
Designer/Engineer	Board Title		
Yusein Ali	SensWear-Debug		
Drawn By	**Document Name**		
Yusein Ali	SensWear-V1R2-Debug-		
Approved By	**Ver.**	**Rev.**	**Date**
Dariush Salami	1.0	R2	20.06.2026

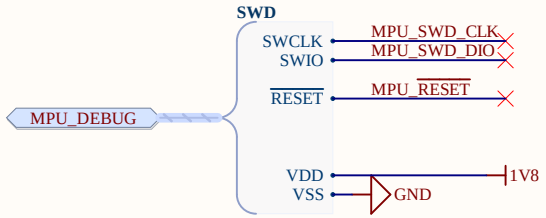


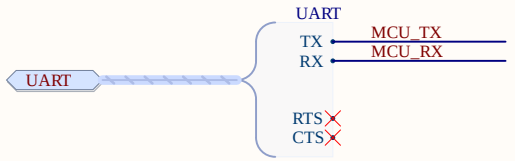
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Designer/Engineer: Yusein Ali	Board Title: SensWear-Debug			
Drawn By: Yusein Ali	Document Name: SensWear_Debug_Main.SchDoc			
Approved By: Dariush Salami	Rev.: R2	Sheet: 1 of 3	Size: A4	
		Date: 20.06.2026		

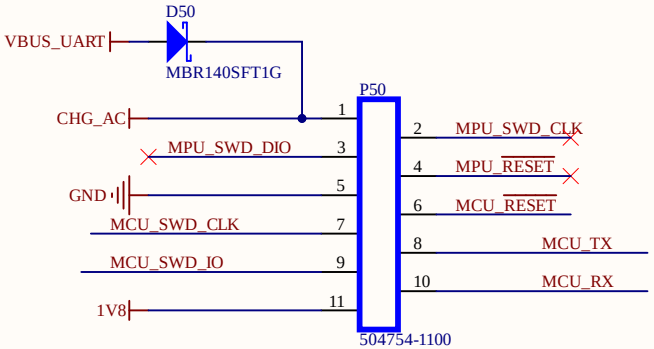
Must mate with the conenctor below





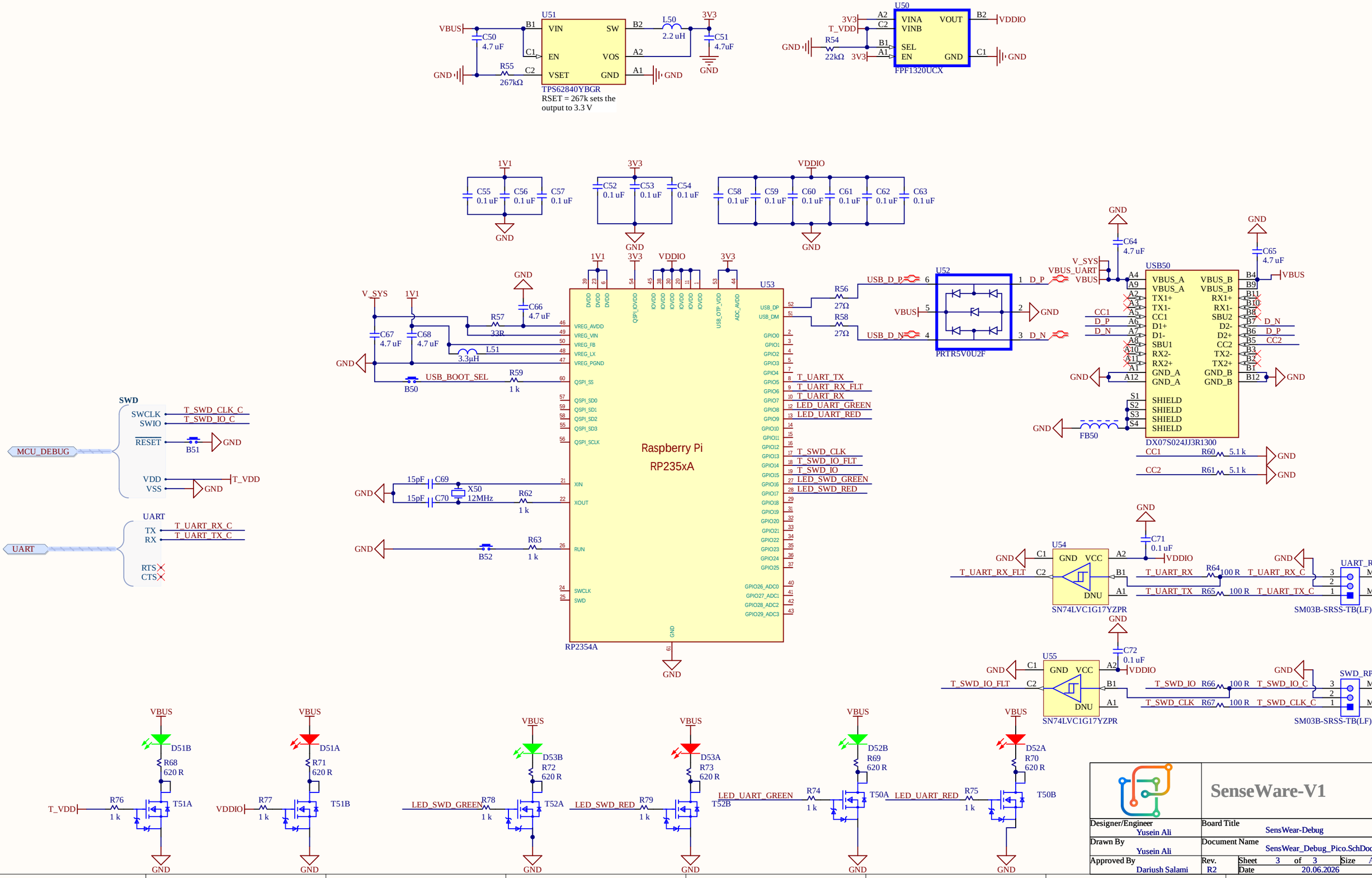






CABLE50
0150310411

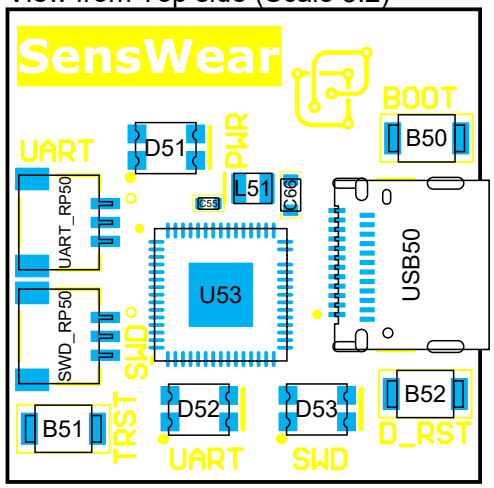
	SenseWare-V1		
Designer/Engineer: Yusein Ali	Board Title: SensWear-Debug		
Drawn By: Yusein Ali	Document Name: SensWear_Debug_DEBUG.SchDoc		
Approved By: Dariush Salami	Rev.: R2	Sheet: 2 of 3	Size: A4
		Date: 20.06.2026	


U51
VIN
SW
EN
VOS
VSET
GND
TPS62840YBGR
RSET = 267k sets the output to 3.3 V
VBUS
C50 4.7 uF
R55 267kΩ
L50 2.2 uH
C51 4.7uF
3V3
U50
VINA
VINB
SEL
EN
VOUT
GND
FPF1320UCX
R54 22kΩ
T_VDD
VDDIO
1V1
C55 0.1 uF
C56 0.1 uF
C57 0.1 uF
C52 0.1 uF
C53 0.1 uF
C54 0.1 uF
C58 0.1 uF
C59 0.1 uF
C60 0.1 uF
C61 0.1 uF
C62 0.1 uF
C63 0.1 uF
U53
Raspberry Pi
RP235xA
RP2354A
V_SYS
C66 4.7 uF
C67 4.7 uF
C68 4.7 uF
R57 33R
L51 3.3µH
USB_BOOT_SEL
B50
R59 1 k
R56 27Ω
R58 27Ω
USB_D_P
USB_D_N
U52
PRTR5V0U2F
C64 4.7 uF
C65 4.7 uF
USB50
DX07S024JJ3R1300
FB50
CC1
CC2
R60 5.1 k
R61 5.1 k
SWD
MCU_DEBUG
SWCLK
SWIO
RESET
B51
VDD
VSS
T_SWD_CLK_C
T_SWD_IO_C
UART
TX
RX
RTS
CTS
T_UART_RX_C
T_UART_TX_C
C69 15pF
C70 15pF
X50 12MHz
R62 1 k
R63 1 k
B52
C71 0.1 uF
C72 0.1 uF
U54
U55
SN74LVC1G17YZPR
R64 100 R
R65 100 R
R66 100 R
R67 100 R
UART_RP50
SWD_RP50
SM03B-SRSS-TB(LF)(SN)
D51B
D51A
D53B
D53A
D52B
D52A
R68 620 R
R71 620 R
R72 620 R
R73 620 R
R69 620 R
R70 620 R
R76 1 k
R77 1 k
R78 1 k
R79 1 k
R74 1 k
R75 1 k
T51A
T51B
T52A
T52B
T50A
T50B
SenseWare-V1
Designer/Engineer
Yusein Ali
Board Title
SensWear-Debug
Drawn By
Yusein Ali
Document Name
SensWear_Debug_Pico.SchDoc
Approved By
Dariush Salami
Rev.
R2
Sheet 3 of 3
Size A3
Date 20.06.2026

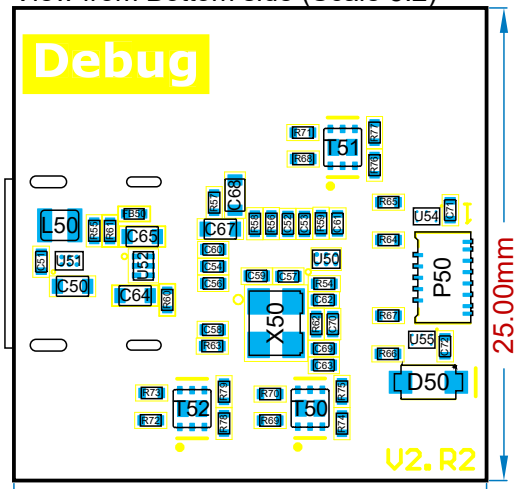
Bill Of Materials

Line #	Designator	Comment	Quantity
1	B50, B51, B52	B3U-1000P	3
2	C50, C64, C65, C66, C67, C68	4.7 uF	6
3	C51	4.7uF	1
4	C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C71, C72	0.1 uF	14
5	C69, C70	15pF	2
6	CABLE50	0150310411	1
7	D50	MBR140SFT1G	1
8	D51, D52, D53	LTST-C155KGJRKT	3
9	FB50	BLM15AG601SN1D	1
10	L50	2.2 uH	1
11	L51	3.3µH	1
12	P50	504754-1100	1
13	R54	22kΩ	1
14	R55	267kΩ	1
15	R56, R58	27Ω	2
16	R57	33R	1
17	R59, R62, R63, R74, R75, R76, R77, R78, R79	1 k	9
18	R60, R61	5.1 k	2
19	R64, R65, R66, R67	100 R	4
20	R68, R69, R70, R71, R72, R73	620 R	6
21	SWD_RP50, UART_RP50	SM03B-SRSS-TB(LF)(SN)	2
22	T50, T51, T52	NTLJD4116NT1G	3
23	U50	FPF1320UCX	1
24	U51	TPS62840YBGR	1
25	U52	PRTR5V0U2F	1
26	U53	RP2354A	1
27	U54, U55	SN74LVC1G17YZPR	2
28	USB50	DX07S024JJ3R1300	1
29	X50	12MHz	1

View from Top side (Scale 5:2)



View from Bottom side (Scale 5:2)



SenseWare-V1

Designer/Engineer	Board Title		
Yusein Ali	SensWear-Debug		
Drawn By	**Document Name**		
Yusein Ali	SensWear-V1R2-Debug-		
Approved By	**Ver.**	**Rev.**	**Date**
Dariush Salami	1.0	R2	20.06.2026